

AAC/MP4 and MP3 Audio Encoding Comparison

General Overview

When you buy and download a song from iTunes, it is encoded in Advanced Audio Coding (AAC) format, which is then containerized in an MP4 file. The reason Apple (as well as Youtube, Nintendo, Sony, and other large tech companies) encode their audio in AAC is because it generally achieves a higher audio quality with the same bit rate as MP3. AAC was originally developed by Dolby, AT&T, Fraunhofer, Sony, Nokia, and other companies as part of the MPEG-2 specification, but was later incorporated into the MPEG-4. The encoding was designed to be the successor of the MP3 format, providing full support of 48 full-bandwidth audio channels in one stream, as well as 16 LFE (low frequency effects). The reason that AAC is able to achieve a higher quality of audio is because the encoding algorithm uses the modified discrete cosine transform (MDCT), which is a sophisticated mathematical algorithm that converts audio signals from the time domain into the frequency domain. The reason the MDCT is so useful for audio encoding is that it allows the encoder to identify and remove frequencies that the human ear cannot differentiate and or hear, decreasing the file size while keeping the audio quality virtually the same.

The reason MP3 requires a higher bitrate to achieve the same audio quality as AAC is that it does not use the newest form of the MDCT when encoding audio signals. Instead, MP3 uses a hybrid algorithm that combines a polyphase filter bank (a processing technique that splits each audio signal into individual frequency bands so that each signal can be compressed independently) and a smaller, much less efficient version of the MDCT to encode. This hybrid algorithm makes the MP3 encoder less precise in terms of how audio data is compressed, leading to loss of quality.

1. Which encoding method offers the best tradeoff between compression ratio and audio quality?

For almost all cases, AAC at 256 kbps is a perfect middle ground between audio quality and file size. The audio quality is almost indistinguishable from lossless, and the file size is about 2MB per minute. As for MP3, 192 kbps is good quality

at approximately 1.5MB per minute, and 320 kbps is considered indistinguishable from lossless with a file size of about 2.4MB per minute. As you can see, AAC can encode the same audio quality for less space, making it more efficient, with MP3 encoding being around 20% larger for virtually the same audio quality.

There are many upsides to MP3, however, including its stability and universal compatibility. MP3 is supported by just about every device, making it ideal for sending and receiving audio files across a wide range of devices. As for its stability, MP3 has been around since the 1990s, and has been a well understood and implemented format in many applications. One of the biggest reasons for developers to use MP3 encoding as opposed to AAC is that AAC is patent-encumbered, which means implementing it costs money. MP3 is currently free and open to implement without any fees, making it the more financially efficient option.

2. Which requires the most processing power for audio encoding?

AAC requires more power to encode than MP3 because the AAC encoding algorithm is more sophisticated and employs more mathematically intense computations than the MP3 encoding algorithm. AAC, however, has the advantage of hardware offloading in terms of how intense the computation is on the CPU. Most new devices have a dedicated Digital Signal Processor (DSP), a component separate from the CPU that is leveraged by device manufacturers to offload the CPU computation for AAC (as well as other format encodings such as image and video) encoding, taking the load off the CPU and in turn making the encoding generally faster than MP3.

3. Which addresses issues of copy protection, and how are these issues addressed?

To protect from piracy, a system called FairPlay (developed by Apple) was applied to AAC that would make it so that no one who was not authorized could play the audio file bought from iTunes. The way they achieved this was by encrypting all files sold on iTunes, and assigning users unique decryption keys for the songs they purchased. That way, if the file was shared with someone without a decryption key (who has not purchased the file), the audio would be a

useless file that could not be played. In 2009, however, Apple removed this feature from AAC encoded files after business negotiations between record labels and Apple, where the record labels wanted their music to be sold on platforms other than iTunes, for which removing the DRM was necessary to achieve. They also argued that the DRM Apple was deploying was hurting sales for the record labels more than piracy was.

MP3 has no DRM or built-in copy protection. The encoding algorithm was designed purely for audio encoding, without proactive measures for anti-piracy or ownership validity. This means that unlike AAC files with FairPlay (or other DRM systems), MP3 files can be freely shared, copied, and distributed after 2017, which is when the patents expired. This feature (or lack thereof) made MP3 the most widely spread audio encoding format for the dissemination of pirated music/audio.

Conclusion

Overall, both MP3 and AAC are very relevant and useful audio encoding formats, each with their own benefits. AAC is the winner in terms of audio quality and space economy, making it a great choice for newer software services; we see the manifestation of these benefits in the implementation of AAC by many software companies that need higher audio quality such as Spotify, Apple Music, and Youtube.

MP3, however, remains a staple in file formats which gives it the benefit of being universal. If you need to share an audio file and the person or company you're sharing the file with must be able to open and make use of it, you have much better luck sharing it in MP3. Ultimately, the 'best' audio format comes down to use case, and both audio formats have established themselves as great choices.